

ES -116

Principles and Applications of Electrical Engineering

Semester II, 2025-26

Frequency shaping with OpAMP based circuits

Candy question before we start

You have built a standard Inverting Amplifier. The non-inverting pin (v_+) is physically wired to ground. You probe the output pin with a highly precise voltmeter, and it reads exactly -10 V.

You look at the Op-Amp's datasheet and see its Open-Loop Gain is **100 dB**.

$$10^5 \text{ V/V}$$

$$\text{V/V} \rightarrow \text{dB} \\ \left(20 \log_{10} \left| \frac{V}{V} \right| \right)$$

1. If you treat this as an "Ideal Op-Amp," what is the assumed voltage at the inverting pin (v_-)?
2. Calculate the *exact* physical voltage sitting at the inverting pin (v_-).
3. Based on your exact calculation, can you assume the inverting terminal to be a virtual ground?

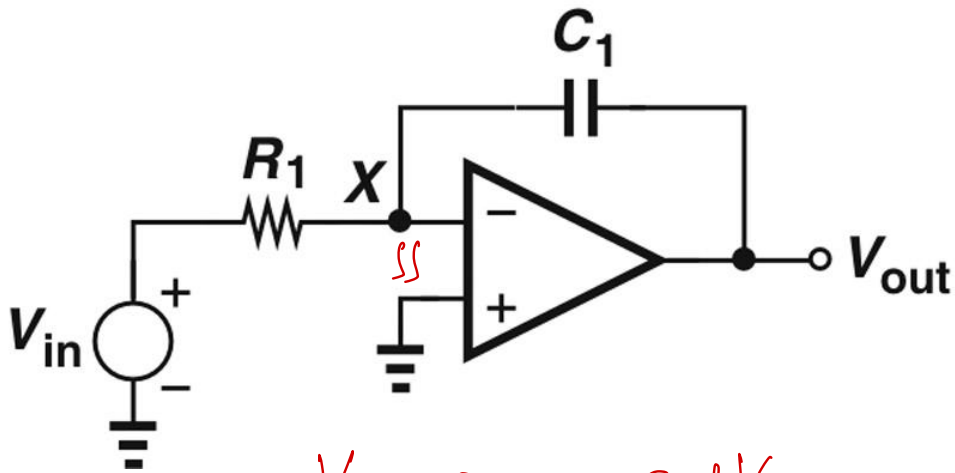
$$V_o = A_o (V^+ - V^-)$$

$$-10V = A_o (0 - V^-)$$

$$V^- = \frac{-10 \times 10}{10^5 \times 10} = -100 \mu V \sim 0$$

Recap

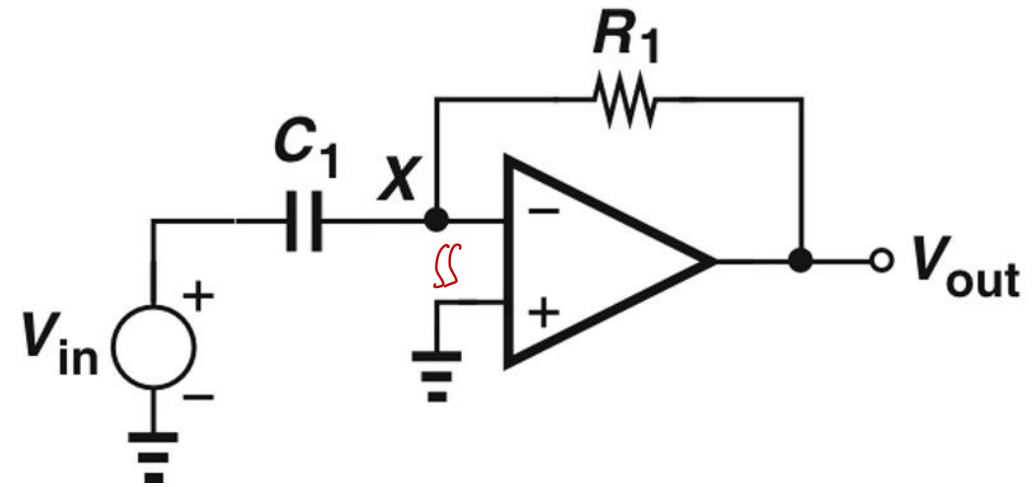
Option 1: $Z_1 = R_1, Z_2 = C_1$



$$\frac{V_{in} - 0}{R_1} = -C_1 \frac{dV_{out}}{dt}$$

$$V_{out} = -\frac{1}{R_1 C_1} \int V_{in} dt$$

Option 2: $Z_1 = C_1, Z_2 = R_1$



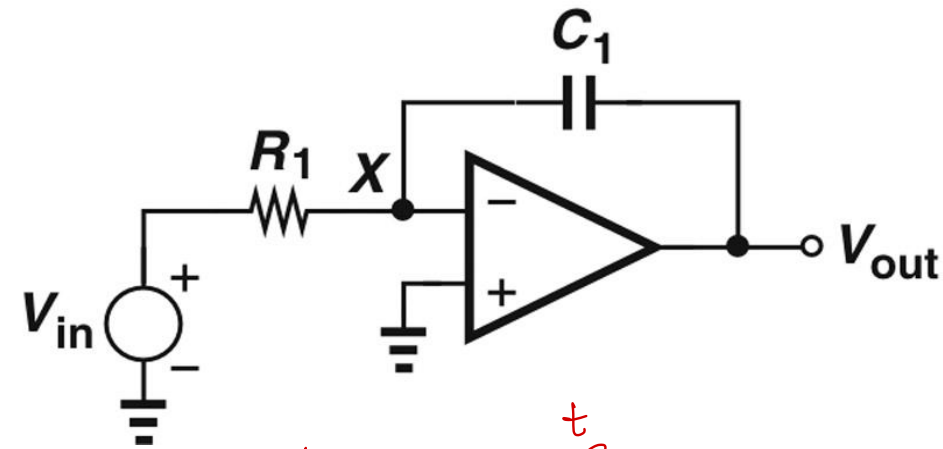
$$C_1 \frac{d(V_{in} - 0)}{dt} = 0 - \frac{V_{out}}{R_1}$$

$$\Rightarrow V_{out} = -R_1 C_1 \frac{dV_{in}}{dt}$$

Functionality in the time-domain

Option 1: $Z_1 = R_1$, $Z_2 = C_1$

Plot V_{out} for a given input.

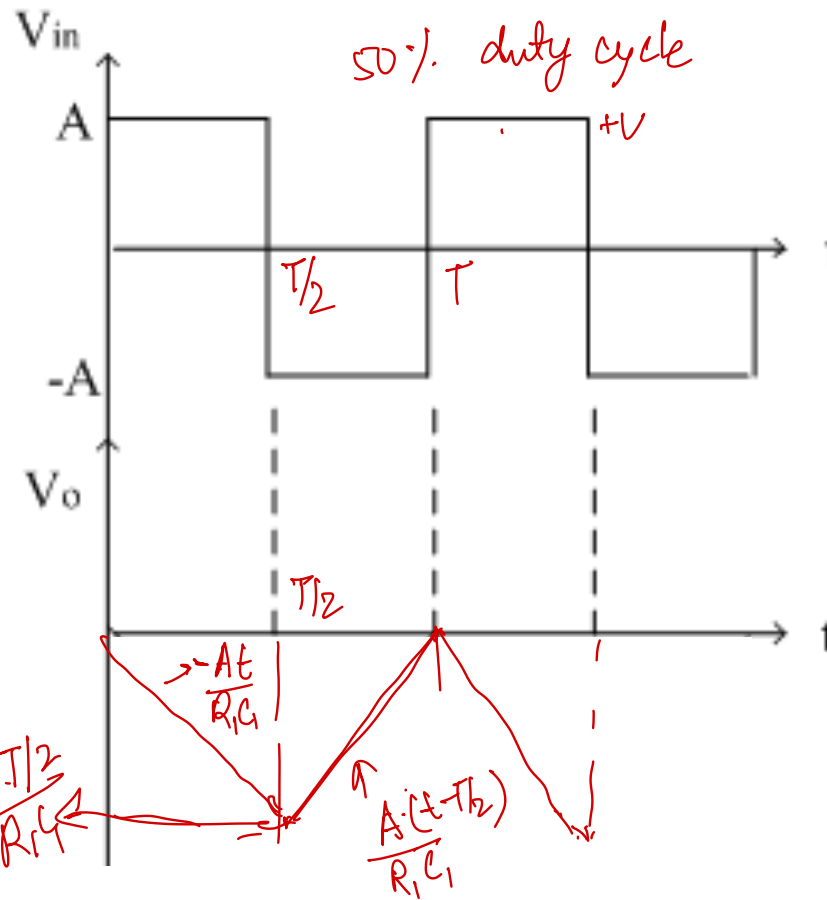


$$V_{out} = -\frac{1}{R_1 C_1} \int_0^t V_{in}(t) dt + V_{out}(0)$$

for $V_{in} = +A$ $0 < t < T/2$

$$V_{out} = -\frac{1}{R_1 C_1} \int_0^t A dt = -\frac{A t}{R_1 C_1}$$

for $V_{in} = -A$ $T/2 < t < T$



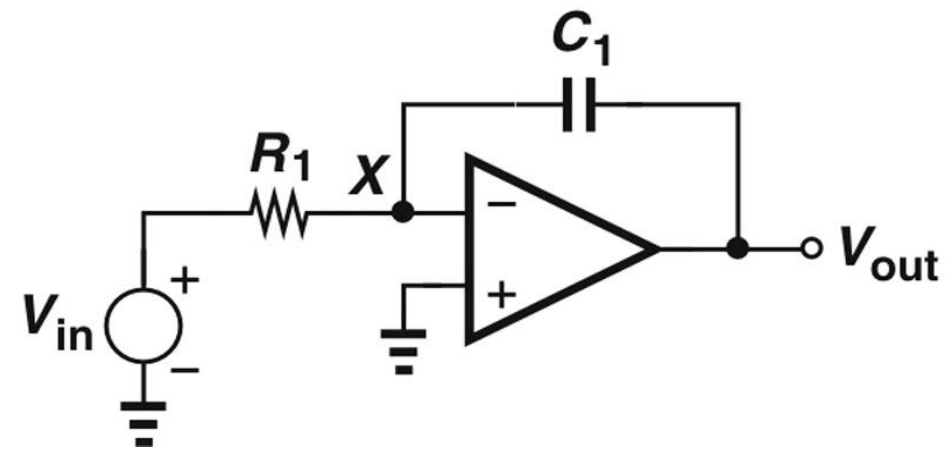
$$V_{in} = -A, \quad T/2 \leq t < T$$

$$V_{out} = -\frac{1}{R_1 C_1} \int_{t_1}^{t_2} V_{in}(t) dt + V_{out}(t_1)$$

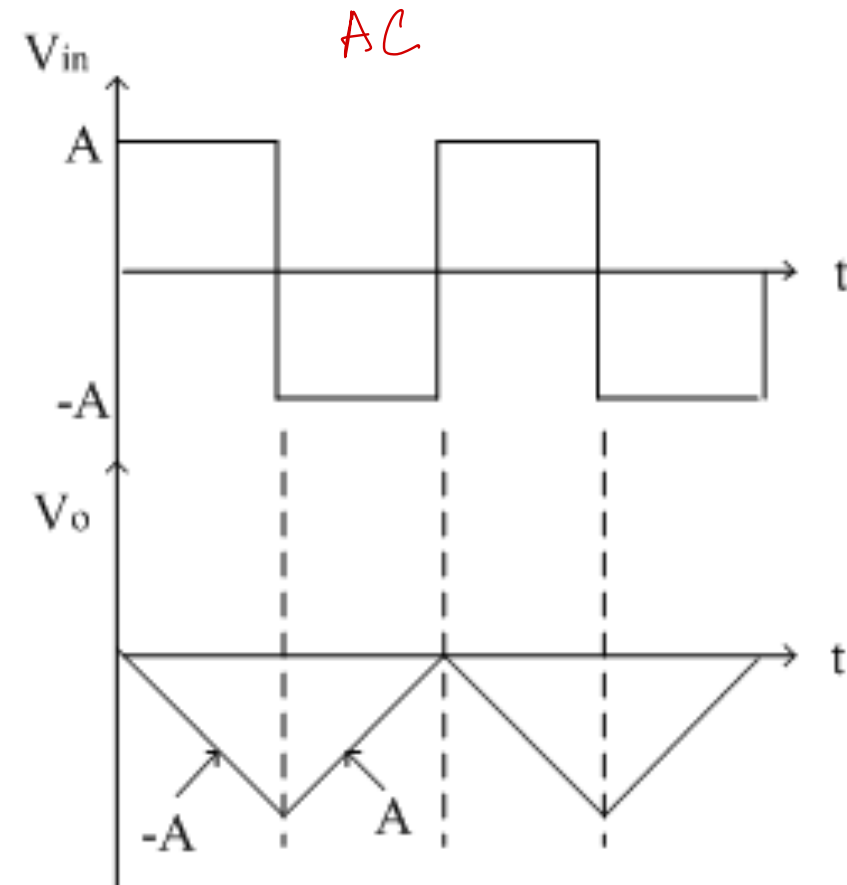
$$= \frac{A}{R_1 C_1} (t - T/2) + V_{out}(T/2)$$

Functionality in the time-domain

Option 1: $Z_1 = R_1$, $Z_2 = C_1$



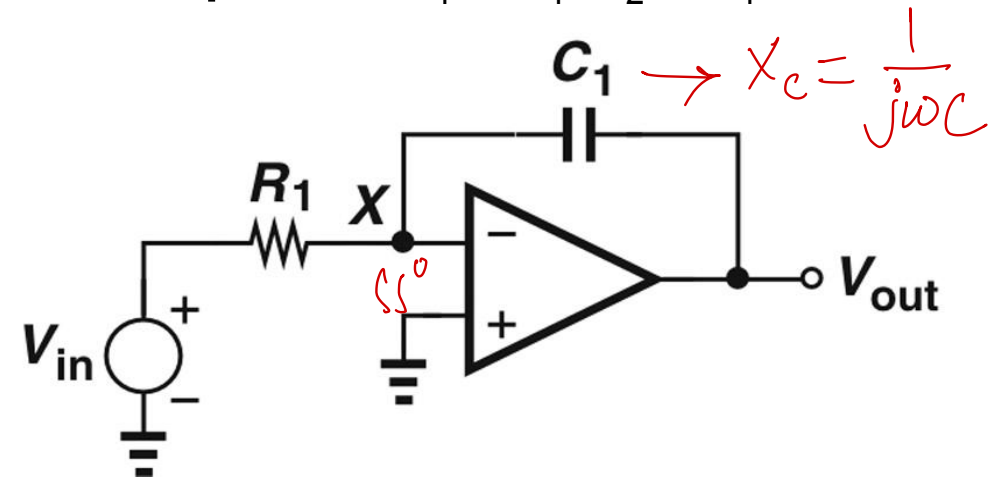
- Can you comment on the frequency spectrum of the input and output signal?
- Physically, sharp (sudden changes) in the time domain would mean higher frequency components, while slower changes means lower frequencies.



Functionality in Frequency domain

Option 1: $Z_1 = R_1$, $Z_2 = C_1$

Plot Gain (V_{out}/V_{in}) with frequency.



$$\frac{V_{in} - 0}{R_1} = \frac{0 - V_{out}}{1/j\omega C_1}$$

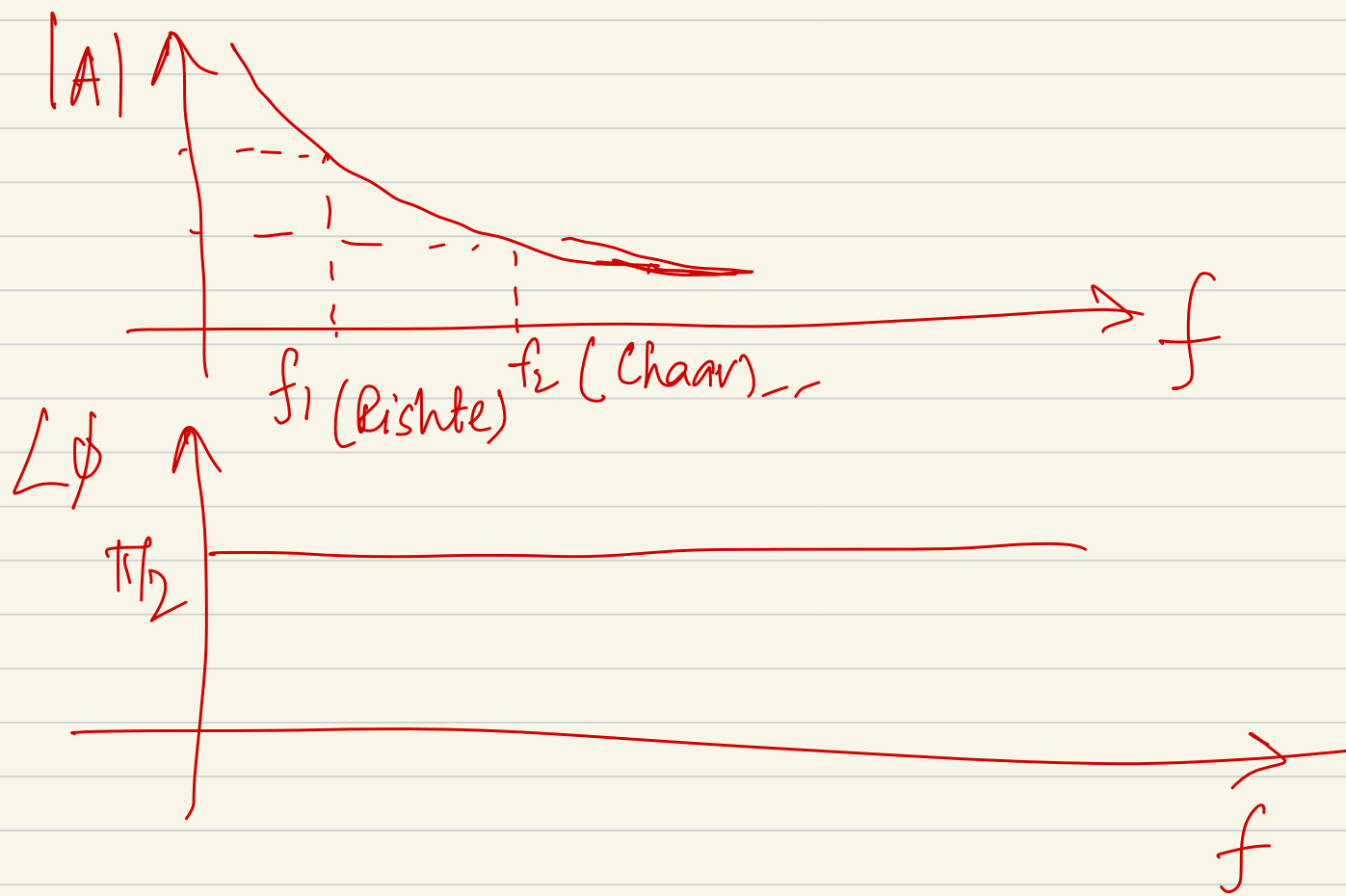
$$V_{out} = -\frac{1}{j\omega C_1 R_1} \cdot V_{in}$$

$$= -\frac{1}{j \cdot 2\pi f R_1 C_1} V_{in}$$

Magnitude & phase

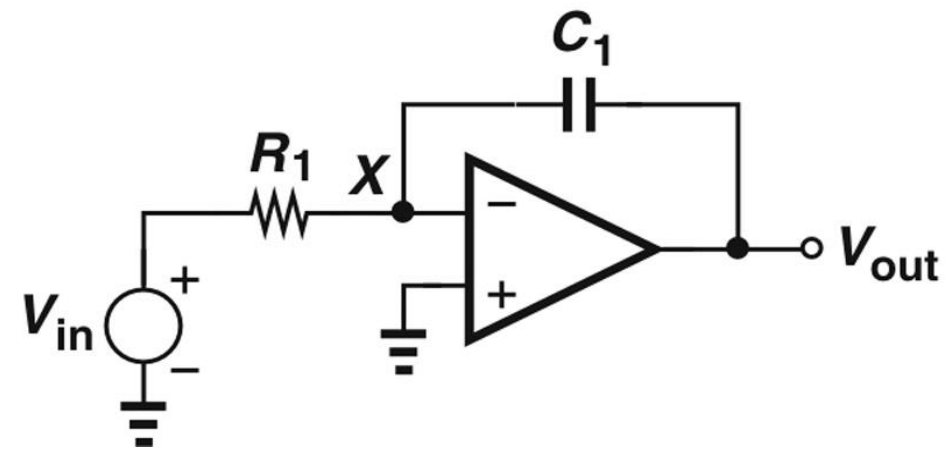
$$|A| = \left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{2\pi f R_1 C_1} \quad \angle \frac{V_{out}}{V_{in}} = \pi/2$$

$$\omega = 2\pi f$$

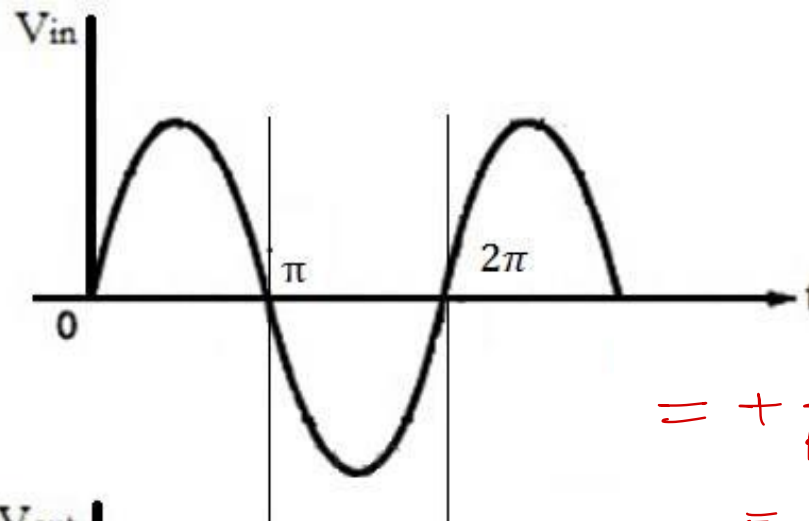


Question

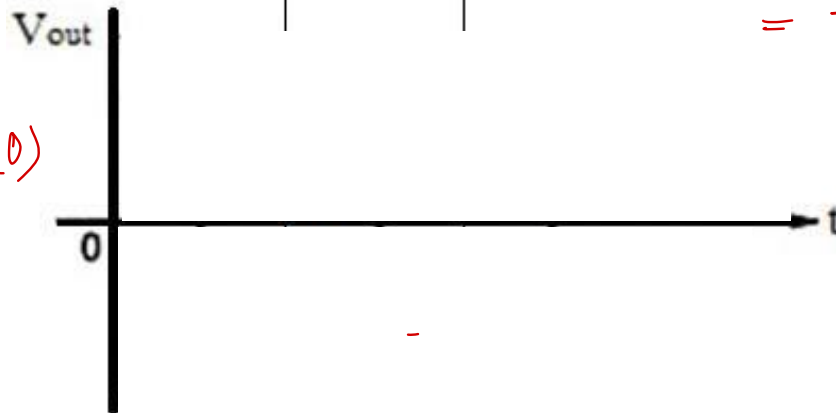
Option 1: $Z_1 = R_1$, $Z_2 = C_1$



Plot V_{out} for a given input and comment on its frequency components.

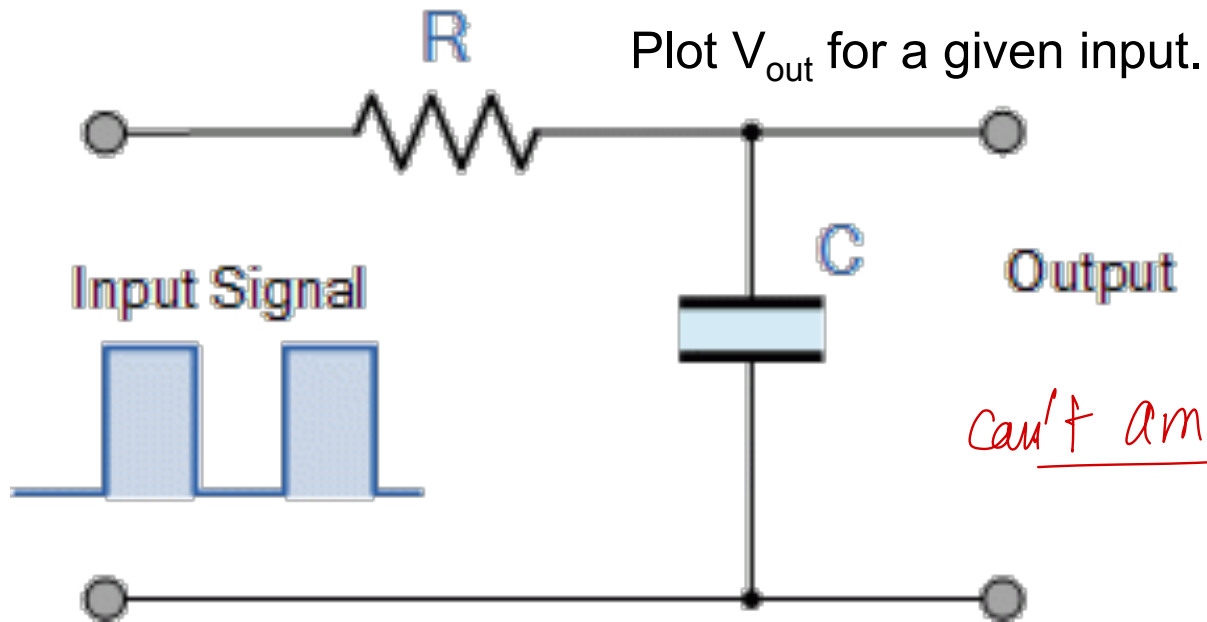


$$\begin{aligned}
 &= -\frac{1}{R_1 C_1} \int V_{in} dt + V(0) \\
 &= -\frac{1}{R_1 C_1} \int_0^t \sin \omega t dt + V(0) \\
 &= -\frac{1}{R_1 C_1} [-\cos \omega t]_0^t + 0
 \end{aligned}$$

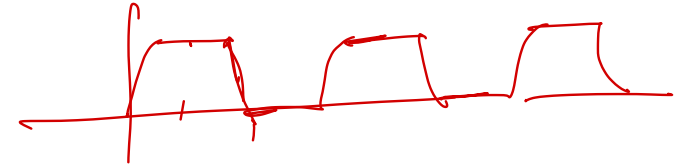


$$\begin{aligned}
 &= +\frac{1}{R_1 C_1} [\cos \omega t - 1] \\
 &= -\frac{1}{R_1 C_1} + \frac{t}{R_1 C_1} \cos \omega t
 \end{aligned}$$

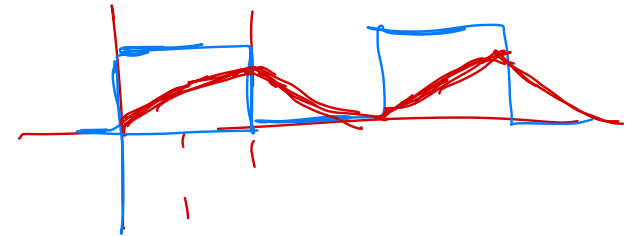
Can't a simple R-C circuit do that?



for smaller C

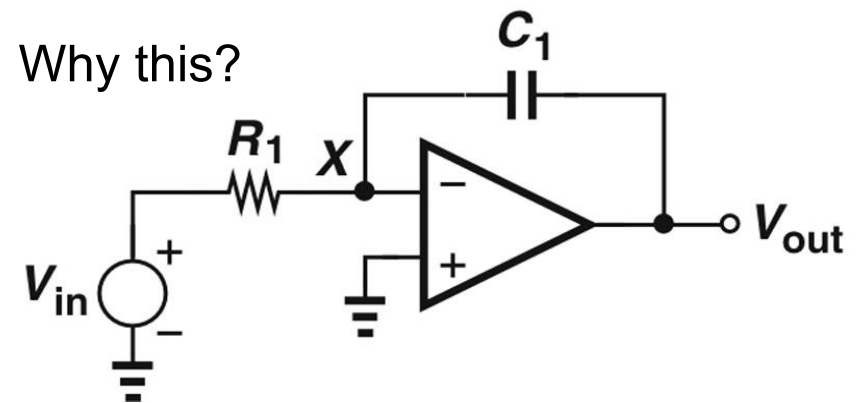


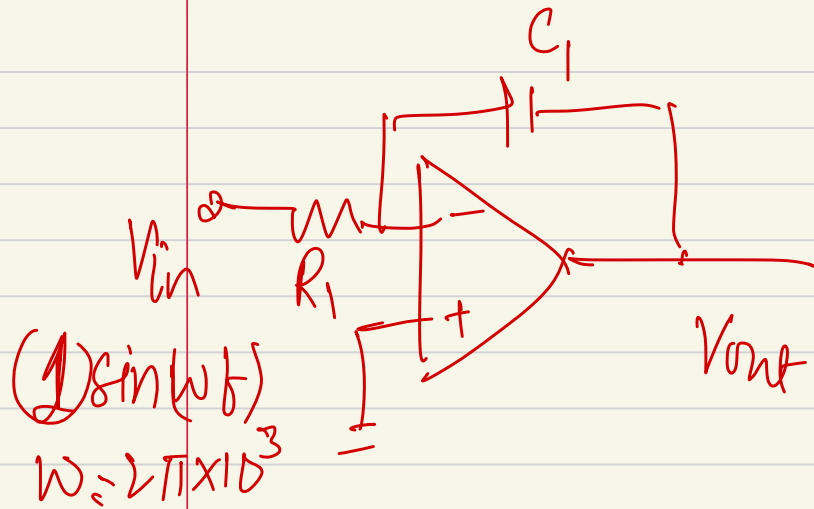
for higher C



can't amplify

Why this?





$$f = 1 \text{ kHz}$$

$$R_1 = 1 \text{ k}\Omega$$

$$C_1 = 1 \mu\text{F}$$

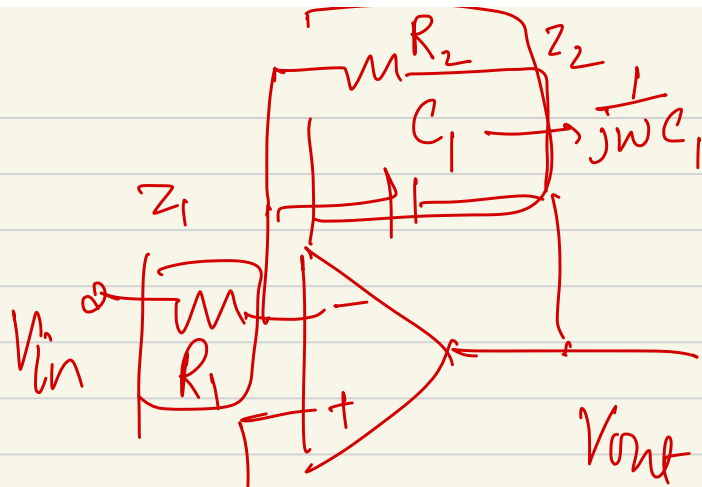
$$R_1 C_1 = 1 \text{ ms}$$

$$V_{out} = -\frac{1}{R_1 C_1} \int V_{in}(t) dt + V_{out}(0) \rightarrow 0$$

$$= -\frac{1}{10^{-3}} \int_0^t \sin(2\pi 10^3 t) dt$$

$$= -\frac{1}{10^{-3}} \left[-\frac{\cos(2\pi 10^3 t)}{2\pi \times 10^3} \right]_0^t$$

$$= \frac{1}{2\pi} [\cos(2\pi 10^3 t) - 1]$$



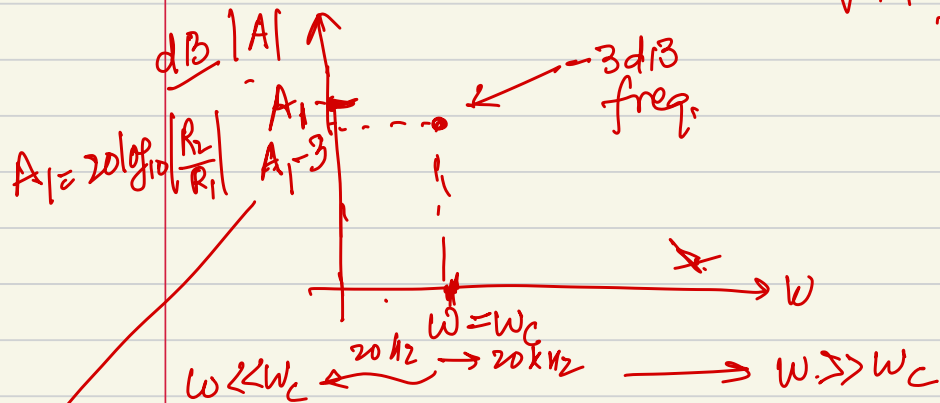
$$V_{out} = -\frac{Z_2}{Z_1} V_{in}$$

$$= -\frac{R_2 \times \frac{1}{j\omega C_1}}{R_2 + \frac{1}{j\omega C_1}} \cdot \frac{1}{R_1} V_{in}$$

$$V_{out} = -\frac{R_2}{R_1} \cdot \frac{V_{in}}{1 + j\omega C_1 R_2}$$

transfer function

$$|A| = \left| \frac{V_{out}}{V_{in}} \right| = \frac{R_2}{R_1} \cdot \frac{1}{\sqrt{1 + \omega^2 C_1^2 R_2^2}} = \frac{R_2}{R_1} \cdot \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$$



$$\omega C_1 R_2 > 1$$

$$\omega C_1 R_2 < 1$$

$$\text{cut off freq} \rightarrow \boxed{\omega_c = \frac{1}{C_1 R_2}}$$

$$A_c = 20 \log_{10} \left| \frac{R_2}{\sqrt{2} R_1} \right|$$

$$= 20 \left[\log_{10} \left| \frac{R_2}{R_1} \right| + \log_{10} \left| \frac{1}{\sqrt{2}} \right| \right]$$

$$= A_1 - 10 \log_{10} 2 = A_1 - 3 \text{ dB}$$

$$\omega / \omega_c \gg 1$$

$$|A| \approx \frac{R_2}{R_1} \cdot \frac{\omega_c}{\omega}$$

$$A_g = 20 \log_{10} \left| \frac{R_2}{R_1} \right| + 20 \log_{10} \left| \frac{\omega_c}{\omega} \right|$$

$$= A_1 + 20 \log_{10} |\omega_c| - 20 \log_{10} |\omega|$$

octave $\rightarrow f_2 = 2f_1$
decade $\rightarrow f_2 = 10f_1$

A_3 ($\omega > \omega_c$)

decreases by -20 dB for each decade of increase in frequency

for $\omega > \omega_c$

$$A_3 = A_1 + 20 \log_{10} |\omega_c| - 20 \log_{10} |\omega|$$

